UNIVERSITY PARTNER

Artificial Intelligence and Machine Learning

(6CS012)

Task 1 Report Question and Answer

Name: Gaurab Rawat

Student Id: 2406775

Group: L6CG18

Supervisor: Mr Shiv Kumar Yadav

Module Leader: Siman Giri

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LONG QUESTIONS

Every day, millions of people use eSewa to send money, pay bills, and shop online. This creates a huge amount of data. The problem is that most of this data has no labels. We do not know in advance which users are like each other, or which transactions are suspicious. This is where unsupervised learning helps. It finds patterns in data on its own, without needing labels. Two areas where this is very useful are grouping customers and catching fraud.

Business Problem 1: Grouping Customers (Customer Segmentation)

The Problem: Users of eSewa have been found to be not uniform. Some students are coming to the university to pay their tuition once a month, others are shop owners and making dozens of payments in a day. Equal treatment of all users is missed by eSewa, which is more advanced in providing better and more personalized services.

The Solution: There exists an algorithm called K-Means Clustering which can be used. This approach considers users' behaviors, including frequency of payment, total transaction amount and timing, as well as types of merchants they pay. It then "clusters" users that are acting in a similar way. Some may comprise individuals that send money to family internationally on a frequent basis, while different groups can consist of people who occasionally send money abroad. When the data features are large in number, some smoothing can be performed using Principal Component Analysis (PCA), which eliminates noise in the data and thus will provide cleaner data for clustering to occur.

How It is Used: The clustering is performed automatically once per week or month. Results are stored in user database. They're used by the marketing team to send offers that are relevant. They are used by them in the product team's planning of what they get built. The support team leverages them to bring the most valuable users prioritized attention. The company can then run tests to evaluate whether these segments make a difference for your sales figures when you create personalized offers from them.

One limitation: K-Means needs to know in advance the number of clusters to form to run the algorithm. The choice of number will have little meaning to the groups if it is not the right number. The algorithm is also based on the assumption of similar size and shape of the groups, and as such this is not always the case with actual user data.

Business Problem 2: Catching Fraud (Anomaly Detection)

The Problem: Some transactions on eSewa are fraudulent. For example, a hacker may take over an account and send large sums of money to unknown people. Or a scammer might make many small payments in quick succession. Fraud is rare, and new fraud methods appear all the time, so there are very few labeled examples to learn from. A model trained only on past fraud may miss new types of attacks.

The Solution: There's a type of neural network that learns what a normal transaction looks like: the Autoencoder. It takes an abbreviated summary of a transaction and attempts to reconstruct the transaction. When a fraudulent transaction comes in, the model cannot rebuild it well because it never learned that pattern. This gives a high reconstruction error, which is a signal that something is wrong. Another option is DBSCAN, an algorithm that marks transactions as outliers if they do not fit into any normal cluster of activity.

How It is used: In real-time operation the Autoencoder is used. Each of the transactions receives a score. A high score indicates that the holder may have verified the suspiciousness of a transaction and then it's marked for manual review or temporarily disables it until it receives confirmation of its authenticity. The system can also be tuned to prevent too many "good" transactions from being rejected.

One limitation: If a fraudulent transaction appears to be quite like normal transactions, then the Autoencoder might not be able to detect it. Moreover, users' daily transactions vary with passage of time, which can lead the model to mark them as fraudulent. That means that the model should be retrained periodically.

SHORT QUESTIONS

Overfitting

What is overfitting and under fitting?

Definitions

Overfitting occurs when too much information goes to your model. It can remember small details, errors and even noise not related to the actual pattern. This gives a model which performs very well on the training data but where it fails very badly on new data it has never seen before. But visualize the student who logs every bit of the answer in his/her practice book word for word in the exam book but can't remember if he/she answered word for word on that same question on the real paper, or if one of the words changes.

Underfitting is what occurs when a model is not complex enough. Does not learn from the training data sufficiently. It has a poor performance in both training and new data. This is like a student that didn't read much and can't give even simple answers.

Both are Issues because of the following:

To be a good model one must generalize. This implies it need to function without issue both on current data and on totally brand-new data. Overfitting done the more particular the model becomes. If underfit, have too broad a definition. Both are of no real use in the "real world". The optimum model falls somewhere between it is complex enough to learn any true pattern that comes along, yet not complex enough to learn any noise.

Examples:

Overfitting: Model that has been trained to classify pictures will memorize the exact lighting and backgrounds of training image. It is trained to be 98% accurate and is 55% accurate on real photographs as it has learned the wrong things.

Underfitting: A model that tries to predict house prices using only the street name will fail. It fails to account for factors such as age of building, number of rooms in the house, neighborhood etc. and so it always predicts wrong.

Neural Network Architecture

The types of data handled by each:

CNNs (Convolutional Neural Networks) are image-structured networks. Image consists of a grid of pixels. The CNN uses small filters to identify the structures, edges, and textures of this grid. Carat's position within the image does not affect the quality of the image. This entire image is filtered with the same filter.

A Recurrent Neural Network (RNN) is designed for Sequences. A sequence is a list of objects in which the order of the objects is significant, such as a list of the words in a sentence or the prices of a stock across time. The RNN sequentially reads the sequence, while keeping track of the information that has been previously seen. This memory is responsible for enabling it to comprehend 'context'.

How Information Flows

The CNN is a one-directional data flow flowing from the input image to the output through several layers. More complex features are detected by each layer than the previous layer. No memory for previous picture.

For an RNN, the information goes through time. With each step, the network changes its memories according to the new word that is sent to it and its existing knowledge. The same, but with the addition of special gates, making it more sophisticated, LTSM. This enables the network to deal with long sequences without losing essential information in the beginning.

To understand how Parameters Are Shared

CNNs share the same filter weights for all location in the image. This implies that it can identify a cat even if its picture is taken in the upper left corner or at the bottom right.

The same weight matrix is used throughout the entire length of the sequence in RNNs. The advantage of this is that the network can accept lengths of sequences as varying as these without having to learn new weights for each length.

Common Training Problems and Fixes

1. For training the network, values of the signals driving the change can get too small to be useful (vanishing gradients) or too large and unstable (exploding gradients). Batch normalization works a little for CNNs, as do skip connections. In the case of RNNs, the LSTM unit and GRU unit can minimize most of the problems of vanishing gradient. Gradient clipping is a way of stopping gradients from going 'out of range'.

2. Drawbacks of a large network: overfitting - easily memorizes training data. To avoid this, we use dropout (randomly disabling some neurons during learning), L2 regularization (where punish large weights), early stopping (as the model may overfit, we stop training earlier) and Data augmentation for image tasks (by flipping or rotating images, new training examples are created).

Real-World Examples

1. CNN: In the medical field, doctors employ CNNs to identify tumors from X-rays and MRI scans. The self-driving cars employ CNNs for recognizing pedestrians and road signs. A CNN is used to unlock face on the phone.

2. In fact, sequence models are used to convert between languages, as done by Google Translate, using the RNN/LSTM architecture. RNN's are used in banks for forecasting a stock price. They are used by voice assistants such as Siri, listening to speech.

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